

Truth Table

- Remas Rawas S24109161
- Rimas Alwassabi S24109600
- Reem Abu alnaja S24209805
- Ward Shaller S24109277

① Task 1

Rain	Wagrid	Dumbledore	KB true
True	True	True	False
True	True	False	False
True	False	True	True
True	False	False	False
False	True	True	False
False	True	False	False
False	False	True	False
False	False	False	False

★ The program prints true because it's the only row that satisfies the statement

② Must be one room, one person, one weapons

Ms. Scarlet must be true because there must be a person and the first two are ruled out

Library must be true because there must be a room & ball room and kitchen are ruled out

Knife must be true because there must be one weapon and revolver and wrench are ruled out.

★ Task 2: Change the knowledge

→ **Change**: delete the last fact **Dumbledore**

→ **Prediction**: the program will print **false**

→ **Results**: the program printed **false**

→ **explanation**: After removing **Dumbledore** the knowledge doesn't prove that **Main** is **true** so the result is **false**

2.2

→ **Delete** the clue **Minerva Gryffindor**

→ **Prediction**: the program will have multiple possible answers so the houses can't be determined uniquely

→ **Results**: the program showed multiple possible house assignment

→ **explanation**: without the **Minerva Gryffindor** clue there isn't enough info to determine one unique solution.

● **Knowledge.add (Symbol ("Severus Gryffindor"))**

New Results

* **Severus** → **Gryffindor**

* **Glendroy** → **Ravenclaw**

* **Pomona** → **Hufflepuff**

* **Minerva** → **Slytherin**

* **Explanation**:

→ The new clue gives **Severus Gryffindor**, which allows the other people to be determined uniquely.

Task 3

```

HACKATHON
> __pycache__
clue.py
harry.py
knights.py
logic.py
mastermind.py
puzzle.py

1 from logic import *
2
3 AKnight = Symbol("A is a Knight")
4 AKnave = Symbol("A is a Knave")
5 BKnight = Symbol("B is a Knight")
6 BKnave = Symbol("B is a Knave")
7
8 knowledge0 = And(
9     Or(AKnight, AKnave),
10    Not(And(AKnight, AKnave)),
11    Implication(AKnight, And(AKnight, AKnave)),
12    Implication(AKnave, Not(And(AKnight, AKnave)))
13 )
14
15 knowledge1 = And(
16    Or(AKnight, AKnave), Not(And(AKnight, AKnave)),
17    Or(BKnight, BKnave), Not(And(BKnight, BKnave)),
18    Implication(AKnight, And(AKnave, BKnave)),
19    Implication(AKnave, Not(And(AKnave, BKnave)))
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```

Welcome Untitled-1 harry.py knights.py

harry.py > ...
1 from logic import *
2
3 rain = Symbol("rain")
4 hagrid = Symbol("hagrid")
5 dumbledore = Symbol("dumbledore")
6
7 knowledge = And(
8 Implication(Not(rain), hagrid),
9 Or(hagrid, dumbledore),
10)
11
12
13 print(model_check(knowledge, rain))
14

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
/usr/bin/env /usr/local/bin/python3 /Users/wardshakir/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 57303 -- /Users/wardshakir/Downloads/AI/hackathon/harry.py
True
wardshakir@wz hackathon % cd /Users/wardshakir/Downloads/AI/hackathon ; /usr/bin/env /usr/local/bin/python3 /Users/wardshakir/.vscode/extensions/ms-python.debugpy-2026.6.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 57325 -- /Users/wardshakir/Downloads/AI/hackathon/harry.py
False
wardshakir@wz hackathon %
```

EXPLORER ... Welcome Untitled-1

1 Open chat (*I), or select a language (*K M), or fill with template to get started.
Start typing to dismiss or don't show this again.

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS zsh + 0 0 ... ^ X

```
wardshakir@wz hack % python3 harry.py; python3 clue.py; python3 puzzle.py; python3 knights.py
True
Traceback (most recent call last):
  File "/Users/wardshakir/Downloads/AI/hack/clue.py", line 1, in <module>
    import termcolor
ModuleNotFoundError: No module named 'termcolor'
GilderoyRavenclaw
PomonaHufflepuff
MinervaGryffindor
HoraceSlytherin
/Library/Frameworks/Python.framework/Versions/3.12/Resources/Python.app/Contents/MacOS/Python: can't open file '
/Users/wardshakir/Downloads/AI/hack/knights.py': [Errno 2] No such file or directory
wardshakir@wz hack %
```

> OUTLINE
> TIMELINE